LAB-III TASK 3 lab Report

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Section B

Title:

Implementation of If Statement in C

Objective:

The objective of this experiment is to understand and implement the `if` statement to check whether a number is positive.

Theory:

The `if` statement is one of the most basic decision-making structures in C programming. It is used to execute a block of code only when a specific condition is true.

Syntax:
```
if (condition)
{
    // statements
}
```

If the condition evaluates to true (non-zero), the statements inside the `if` block are executed. If the condition is false (zero), the block is skipped.

In this task, we use the condition `num > 0` to determine whether a number is positive.

Program (with comments):

```
#include <stdio.h>   // Standard input-output library

int main()
{
    int num; // Declare integer variable

    printf("Enter an integer: ");  // Prompt user to enter a number
    scanf("%d", &num);             // Store user input

    if (num > 0)                   // Check if number is greater than zero
    {
        printf("The number is positive.\n"); // Display message if condition
is true
    }

    return 0; // End of program
}
```

Detailed Explanation:

The program begins by including the `<stdio.h>` library, which provides input and output functions.

A variable `num` is declared to store the user's input. The program asks the user to enter an integer and stores it using `scanf`.

The `if` condition `num > 0` checks whether the entered number is positive:

- If the condition is true, the program prints that the number is positive.
- If the condition is false (i.e., the number is zero or negative), the program does nothing because there is no `else` block.

This demonstrates how a simple `if` statement controls program flow based on a condition.

Output (Example):

Enter an integer: 7
The number is positive.

Conclusion:

The `if` statement allows execution of specific code only when a condition is satisfied, making it essential for decision-making in programming.